MITESH UPADHYAY

SUMMARY

- Innovative and results-driven AI Architect with over 13 years of experience in designing and delivering cutting-edge solutions.
- Led end-to-end data science initiatives, from problem definition and data acquisition to model deployment and post-production monitoring —driving measurable business impact through advanced statistical modeling, machine learning, and generative AI techniques.
- Directed cross-functional teams of data scientists, ML engineers, and analysts, providing technical mentorship on experimentation, feature engineering, RAG/LLM workflows, and model governance while ensuring alignment with product, engineering, and business stakeholders.
- Expert in Generative AI and Agentic AI solutions across diverse industries. Adept at architecting intelligent ecosystems that combine RAG (Retrieval-Augmented Generation), FineTunning, autonomous multi-agent frameworks, MCP (Modular/Managed Component Pipelines) and A2A(Agent to Agent) protocol to knowledge orchestration, and decision intelligence.
- Proven expertise in leveraging LLM orchestration frameworks, prompt engineering, and AI agent collaboration to build high-performing systems. Skilled in integrating cloud-native AI stacks (AWS, GCP) with vector databases, embeddings for adaptive reasoning and contextual intelligence, along with leveraging AWS Bedrock, Google Vertex AI, and Autogen frameworks.
- Expertise in MLOps, LLMOps, DevOps, Inference and Training Data Pipelines/Management/Ingestion, Automation of data movement, transformation, quality and integrity verification , production drift monitoring and developing end to end CI/CD pipelines, developing & deploying scalable models, orchestration, scheduling, automation, ML Operations.
- Expertise in generative AI models (like GPT4, Claude 3, Turbo3.5 , DeepSeek, Llama etc). Also exposure to integrating LLMs(Open AI GPT) using API endpoints.
- Conduct exploratory data analysis (EDA) to uncover trends, patterns, and relationships in data, overseeing data acquisition, analysis, and generating valuable insights.
- Thorough understanding of prompt engineering; implementation exposure to LLM agents like Lang Chain & vector databases, RAG implementation .
- Knowledge of the latest techniques in this space such as LORA adaptors, PEFT, quantization, fine tuning LLMs including BERT and GPT based.

CONTACT

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SKILLS

- Python
- MLOps
- Docker
- Deep Learning
- SQL
- Terraform
- Fast API
- Open AI
- Flask, Streamlit
- Java
- Sklearn
- Tensorflow
- Keras
- Sagemaker
- Amazon Web Services
- Bedrock
- PyTorch

- Use Python and related libraries (Numpy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch etc.) for data analysis, model training, visualize and prediction.
- Hands on knowledge of Feature Store, MLflow, model registry, microservices architecture, Flask, Docker, Kubernetes.
- Showcased expertise in infrastructure as code, specifically utilizing Terraform to enhance system scalability and ensure optimal availability.
- Well-versed in advanced NLP concepts such as RNN, LSTM RNN, and Transformer, BERT contributing to enhanced data preprocessing and sentiment analysis.
- Thorough understanding of Deep Learning , Neural Network ,Machine learning concepts around regression, classification, clustering, dimensionality reduction, NLP, CNN, RNN etc.
- Experience in Data Cataloging, Data Governance and Data Security.
- Experience with Databricks, Data Warehouses, Data Lakes, also with scripting in SQL - extracting large data sets and design of ETL flows.
- Proficiency in designing cloud native architecture & programming languages such as Python, Java , React JS and proficient in web and API development.

CERTIFICATIONS

- AWS Certified Machine Learning Speciality
- AWS Certified Solution Architect Associate
- AWS Certified Developer Associate
- Terraform Certified Associate
- Python PCEP-301

EXPERIENCE

Senior Solution Architect - AI, **12/2025 - 03/2026**
Coforge Ltd - Pune, India

- Engage with business stakeholders to understand client requirements, pain points, and opportunities for AI-driven solutions.
- Translate business problems into scalable AI/ML and Generative AI solution architectures.
- Collaborate with sales and consulting teams during RFPs, proposals, and client presentations.
- Present architecture designs, AI strategies, and technical solutions to technical and non-technical stakeholders.
- Conduct workshops, solution walkthroughs, and architecture reviews.
- Act as a trusted AI advisor for clients by recommending future AI initiatives and transformation strategies.
- Engage with clients to identify AI opportunities and translate business needs into scalable AI/GenAI architectures.

- Support pre-sales activities including proposals, solution design, and technical presentations.

Technical Lead AI/ML, 02/2025 - 11/2025
Ankercloud Technologies Pvt Ltd - Pune, India

- Architect and implement end-to-end Generative AI solutions using AWS Bedrock, Google Vertex AI, and custom LLM frameworks for enterprise-scale applications.
- Design and deploy RAG (Retrieval-Augmented Generation) pipelines integrating vector databases (e.g., Pinecone, Weaviate, FAISS, OpenSearch) with structured/unstructured enterprise data sources.
- Build and orchestrate Agentic AI systems leveraging Autogen, LangGraph, and CrewAI to enable multi-agent collaboration, autonomous reasoning, and adaptive task execution.
- Develop and manage MCP (Modular / Multi-Component Pipelines) for large-scale AI workflows, ensuring modularity, scalability, and cross-agent communication.
- Evaluate, fine-tune, and operationalize LLMs (Claude, Titan, Gemini, Mistral, etc.) across AWS Bedrock and Vertex AI platforms for domain-specific intelligence.
- Implement prompt engineering and prompt-chaining strategies to optimize LLM reasoning, contextual awareness, and response accuracy.
- Define AI governance, observability, and performance monitoring frameworks using tools like LangFuse, Weights & Biases, and Bedrock Model Evaluation.
- Design multi-modal and multi-agent AI systems combining text, image, and video intelligence using Bedrock and Vertex AI generative services.
- Evaluate and integrate open-source AI frameworks (LangChain, LlamaIndex, Hugging Face) with cloud GenAI services to enhance flexibility and cost efficiency.
- Implement MLOps frameworks for LLMOps, enabling automated model versioning, experiment tracking, and CI/CD pipelines using SageMaker Pipelines, MLFlow, Airflow and Kubeflow.
- Containerize and orchestrate AI workloads using Docker, Kubernetes (ECS), and Terraform for scalable, portable deployments.
- Experience in web Scrapping using Tavily , DuckDuckGo, SerperAPI.

Solution Architect , 04/2021 - 01/2025
John Deere India Pvt Ltd - Pune

- Simulating, designing, developing & deploying computationally complex and practical data; building and delivering comprehensive data strategy roadmaps; ensuring final deliverables are of the highest quality.
- Explored datasets to identify and address data transformation and quality requirements tailored to specific applications.
- Applied regression and classification techniques extensively to derive meaningful predictions and insights.

- Managed data acquisition and conducted in-depth analysis to extract actionable insights across diverse domains.
- Leverage prompt engineering and tuning strategies for improving LLM reasoning, precision, and contextual alignment with enterprise knowledge.
- Implement feedback loops, evaluation metrics, and model performance monitoring using LangFuse, Weights & Biases, and RAGAS for continuous improvement.
- Develop data preprocessing, embedding generation, and chunking workflows for high-quality knowledge ingestion and retrieval optimization.
- Automate model deployment, retraining, and scaling through MLOps & AIOps pipelines built with SageMaker Pipelines, Vertex AI Pipelines, and CI/CD frameworks (GitHub Actions, Jenkins).
- Implement and maintain AI microservices and APIs using FastAPI, Flask ensuring scalable and low-latency communication across AI systems.

Senior Software Engineer, **08/2018 - 04/2021**
Tech Mahindra - Pune, India

- Developed statistical models and applied advanced machine learning techniques to enhance data-driven decision-making.
- Orchestrated the deployment and optimization of machine learning and NLP solutions on web-based applications hosted on AWS.
- Working with stakeholders to define business & systems requirements for new information technologies.
- Evaluating data models and studying forecasts & demand; implementing databases, and writing stored procedures, analyzing content and quality of databases and recommending data management procedures.
- Building and maintaining ML production pipelines.
- Evaluated model performance using appropriate metrics and fine-tune hyperparameters to achieve optimal results.
- Played a pivotal role in seamless integration across multiple teams, fostering collaboration and synergy.
- Serves as a technical peer and a individual contributor for junior-level colleagues ongoing technical training, peer review etc.

Senior Software Engineer , **02/2017 - 08/2018**
Cybage Software - Pune, India

Associate Consultant, **03/2015 - 02/2017**
Capgemini Ind Pvt. Ltd - Mumbai, India

Software Engineer , **11/2013 - 03/2015**
3i Infotech - Mumbai, India

Assistant System Engineer, **10/2012 - 10/2013**
Tata Consultancy Service - Mumbai, India

SPECIALIZATION

- Post Graduation in Machine Learning & AI (approved by AICTE and recognized by the UGC & Program is a WES recognized)

PROJECTS UNDERTAKEN

1) MASO - MutliAgentic Service Orchestrator - (Club Mahindra)

· Built a Multi-agent AI system that autonomously interprets customer intents, validates policy constraints, and orchestrates backend actions across multiple systems — improving response accuracy and reducing failed bookings.

- Supervisor Agent (Router): Entry point that classifies intent and delegates to specialized collaborator agents.
- Collaborator Agents:
  Enquiry Agent – Handles general queries, retrieves hotel & policy details via Bedrock Knowledge Base (RAG-based, OpenSearch embeddings over S3 documents).
  Booking Agent – Validates credit-day balance, checks room availability, and performs soft bookings via DynamoDB integration.
  Support Agent – Manages customer complaints, logs CRM tickets, and routes human callback requests.

Each agent uses dedicated Action Groups (Bedrock Tools) to execute domain-specific tasks — such as querying DynamoDB, fetching from OpenSearch, or posting to the ticketing system. The Bedrock Knowledge Base provides context-grounded responses for policies and FAQs, ensuring factual and compliant interactions.

· Key Features & Flow:

- Intent classification and routing through Supervisor Agent.
- Policy-aware vacation recommendations and plan validation.
- Real-time balance check and soft booking with a 6-hour payment window.
- Automated complaint logging with ETA tracking and human escalation.
- Contextual retrieval from RAG-powered Bedrock Knowledge Base with OpenSearch embeddings.
- All interactions governed by Bedrock Guardrails (PII redaction, policy compliance).

2) MIRA - Medical Intelligence & Reasoning Agent - (CIPLA Ltd)

·Designed and developed an advanced Generative AI–powered Voice Chatbot for CIPLA, a leading pharmaceutical company, to enable seamless multilingual interactions with healthcare professionals and patients. The system was fine-tuned on domain-specific medical data, empowering it to understand and respond accurately to complex clinical and pharmaceutical queries through both voice and text modalities.

Key Features & Highlights:

- Multimodal Architecture:
  Integrated image, text, and voice understanding using Vision-Language Models (VLMs) to interpret product labels, prescriptions, and package

images along with text-based queries.
Models leveraged included Google's Gemini Pro (for multimodal reasoning) and AWS Bedrock Titan Embeddings for semantic search and context retrieval.

- **Voice-Enabled Conversational Layer:**
  Implemented Google Speech-to-Text API for multilingual voice input and Google Text-to-Speech for natural voice responses. Supported 10+ Indian and global languages with context-aware tone and pronunciation adaptation.
- **Domain-Specific Fine-Tuning:**
  Fine-tuned LLM models such as Falcon, Llama 3, and Claude on proprietary CIPLA medical datasets using LoRA (Low-Rank Adaptation) and Parameter-Efficient Fine-Tuning (PEFT) techniques to optimize for accuracy, reduced hallucination, and domain relevance.
  Training and inference were orchestrated via Amazon SageMaker.
- **RAG (Retrieval-Augmented Generation) Framework:**
  Built a RAG pipeline to fetch up-to-date drug information, dosage guidelines, and regulatory documents.
  Vectorization: AWS Bedrock embeddings (Titan G1)
  Storage & Retrieval: AWS OpenSearch
  Orchestration: LangChain and FastAPI-based backend with context caching
- **Multilingual NLP Layer:**
  Utilized Google Translation API for translation and transliteration support across multiple Indian languages (Hindi, Marathi, Tamil, Bengali, etc.) ensuring inclusivity and accessibility.
- 3) **LearnSentra — Multi-Agent Adaptive Learning & Assessment Platform (MCP + A2A) - (BYJU's)**

· Architected LearnSentra, a multi-agent education intelligence layer for the BYJU'S app that delivers personalized tutoring, adaptive practice, and mastery tracking across K-12 (CBSE/ICSE/State boards) and JEE/NEET prep.

- **Tutor Agent** – Conversational explanations aligned to BYJU'S videos and pedagogy; multilingual (English + Hindi + regional languages).
- **Assessment Agent** – Adaptive quizzes, past-year question drills, concept mastery scoring, item-response theory (IRT) calibration.
- **Content Curation Agent** – Maps syllabus → concepts → BYJU'S video capsules → practice sets; ensures grade/board alignment.
- **Progress & Recommendation Agent** – Detects weak concepts, sequences next best activity (video → practice → revision).
  The system uses **MCP (Model Context Protocol)** to safely invoke BYJU'S internal tools/content APIs and **A2A (Agent-to-Agent)** collaboration (LangGraph/CrewAI) to coordinate tutoring, assessment, and recommendations in real time—online and low-bandwidth/offline-first.

Results

- **+38%** improvement in *concept mastery* (2-week pilot, n≈5k learners across Grade 8–10 Math).
- **−52%** reduction in *time-to-understand* target concepts when Tutor→Assessment→Video loop triggered automatically.

- **+31%** uplift in *7-day learning streaks* via nudge scheduling and gamified XP.

LANGUAGES

Gujarati: First Language

English: B2

Upper Intermediate (B2)

Hindi: B2

Upper Intermediate (B2)

EDUCATION

Master of Computer Applications, **Computer Science And Programming**, **06/2012**
Shri Vaishnav Institute of Management - Indore